

Deep Learning Framework Facilitated Design, Development, and Analysis of Chronic Diarrhea Syndrome Based Dog Emotion Identification and Classification System Utilizing Hybrid Filter Models

Shivaprasad, Kusumadhara S, Ujwal U J

Abstract— Dogs are the most preferred domestic pets generally adopted by the human beings, as they stay kind and loyal towards their owners than any other domestic pets. To maintain cordial relationship with the pets it is necessary to understand pets' emotion more precisely. Accurate emotion recognition of dogs can help in enhancing the owner-pet interaction, communication and overall well-being of the pet. Emotion recognition in dogs is a challenging task due to the complexity involved in their physiological, behavioural and vocal characteristics. As a result, there is a strong need to design an automated emotion recognition system to prevent mistreatment of pets. This paper aims to design an automated emotion recognition system based on Hybrid filter (Wavelet/LMS with IIR-MFCC) and Deep learning approaches. The different dog vocalizations analysed for emotion classification in this study include other animal voice, dog-barking, dog-grunting, and dog-growling. To enhance the classification performance, the raw audio files underwent preprocessing based on cascaded DWT- IIR filter and LMS-IIR filter. The pre-processed data was subsequently applied to the MFCC system to extract features, which served as the input to the classification model. Classification results for the various dog-barks using several Machine Learning and Deep Learning approaches were recorded. Deep learning models namely Alex net-CNN and Lenet-5 CNN produced far superior results on the dog-emotion dataset. Accuracy of 87.5 and 85 were achieved on the testing dataset using these models.

Index Terms— Alex-net, Lenet-5, CNN, Hybrid filter

I. INTRODUCTION

HUMANS often choose dogs as their top preference for domestic pet adoption. Other common pets include cats,

cows, rabbits, and more. Cows, in particular, are adopted for practical reasons, as they provide valuable products like milk and dung, which are essential for various human needs. However, caring for a cow is demanding, requiring at least two to three hours of daily attention. On the other hand, pets like cats and rabbits offer fewer direct benefits to humans but are easier to manage.

Dog emotions can be investigated by means of body language pose or by its voice cry. Often it is difficult to manually investigate the emotions of dog with extreme effectiveness, sometimes dog owner may go wrong identifying its emotions.

Chronic diarrhea syndrome in dogs is typically caused by bacterial contamination in food or water. In most cases, the condition does not persist for long. However, if it continues for years, pet owners may face added challenges in managing their pets. These dogs tend to bark more frequently throughout the day, requiring constant attention. Elderly owners, in particular, may find this routine especially demanding. Hence by introducing automated voice recognition and classification model, human misinterpretation error can be reduced significantly.

A. Dog emotions understanding by its body language

Tail fluttering: This particular body language posture conveys the dog's joyful mood to its owner. It is often observed when reuniting with the dog after an extended absence; the dog expresses happiness by wagging its tail.

Bristling fur: This sign gesture suggests that the dog may be experiencing a range of emotions, including being upset, stressed, excited, or interested, depending on the specific circumstances. Deciphering the exact issue can be challenging for humans due to the ambiguity of the signal and the dog's diverse emotional states.

Posture: When a dog hunches towards the ground, it signifies a sign of fear or a submissive state. This behavior is often triggered when the dog anticipates punishment for mistakes or when confronted by more dominant dogs barking, causing fear. Another observable posture is when a dog rolls onto its back, exposing its belly. This gesture indicates that the dog is in a relaxed and playful mood, expressing a desire for interaction and playfulness.

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Shivaprasad is a Research Scholar at KVGCE Sullia, affiliated to Visvesvaraya Technological University, Belagavi 590018 INDIA (Corresponding Author, phone: 917760492838, e-mail: shivaraol01@gmail.com).

Kusumadhara S is a Professor and Head of the Department of Electronics and Communication Engineering at KVGCE Sullia, affiliated to Visvesvaraya Technological University, Belagavi 590018 INDIA (e-mail: kusumadhara5@gmail.com).

Ujwal U J is a Professor, Faculty of Computer Science and Engineering, Registrar (Eval), Visvesvaraya Technological University, Belagavi 590018 INDIA (e-mail: ujwalu@yahoo.com).

Facial demeanor: A dog's eyes can convey significant information about its emotional state. Soft eyes typically indicate that the dog is calm or happy. On the other hand, hard eyes or prolonged staring can suggest that the dog may be feeling aggressive, often serving as a sign of a potential threat. Monitoring a dog's eye expressions is a valuable means of understanding its current mood and potential behavior [1].

B. Dog emotions understanding by various means of barks

Normal bark: This is normal bark, used to communicate with owner for the various reasons which includes take it for a walk, it is hungry, get your attention etc. No major risk during this sign.

Grunt: This bark sign indicates the dog is stressed, uncomfortable or tied alone in the room. It is state which needs to be resolved; appropriate actions need to be taken by the owner

Growl: This bark sign indicates the dog is really angry. Situation arises when your dog starts quarrelling with other dogs or if it observes other animals(stranger) approaching it. Necessary immediate actions need to be taken by the owner under such circumstance [2].

The dog grunt and growl bark sounds are acquired from a dog suffering from chronic diarrhea.

II. LITERATURE REVIEW

Ferres K et al. [3] introduced a system for smartly identifying various dog emotions, including “anger”, “fear”, “happiness”, and “relaxation”. The system's learning process relied on a historically-trained ML model that used smart pose estimation to distinguish between canine “emotional” states. A “query” dog key point detection model was developed using the DeepLabCut framework for animal key point detection training. The model was trained on “13,809” annotated dog images and is capable of estimating the coordinates of “24” different dog body part key points. The system achieved an accuracy of “60%” to “70%”, surpassing human ability to identify dog emotions.

Cleary Michelle et al. [4] conducted a review of recent literature on owner-perpetrated animal cruelty, focusing on its prevalence, underlying motivations, and consequences. The study particularly explored the link between animal abuse and intimate partner violence (IPV), emphasizing the impact on both human victims and their pets.

Scoresby KJ et al. [5] examined recent peer-reviewed studies to gain a deeper understanding of what is currently known about the connection between pet ownership and mental health. The findings suggest that owning a pet typically has a more positive than negative effect on mental well-being, emphasizing the advantages of having pets in people's lives.

Maja Scheider et al. [6] proposed a robust system for dog voice analysis and classification with audio data collected using a microphone connected to a RPi 3 Model B running the Raspbian operating system. The system captured recordings at a sampling rate of 48 kHz over a 30-minute duration. The dataset included barks from a specific dog, as well as vocalizations from other animals. For analysis, audio signals were divided into frames of 21 milliseconds, and both temporal and spectral features were extracted, resulting

in a feature vector of up to 8,239 dimensions. LDA was employed to reduce the feature dimensionality to 51. Among the classifiers evaluated, SVM and RF demonstrated superior performance, with the SVM achieving a low false positive rate of 2% and a sensitivity of 89%.

Sibiryakova OV et al. [7] conducted a study to examine the frequency and temporal characteristics of whines from 20 adult companion dogs (“9 males”, “11 females”), with body masses ranging from “3.5” to “70.0” kg, across “16” breeds. A total of 824 whines (26-71 per dog) were recorded in conditioned begging situations created by the owners. The frequency content of the whines was analyzed using the FFT tool, and an ANOVA classifier was applied to identify different classes of dog whines.

Yesy Diah Rosita et al. [8] developed a system to classify infant cries based on their vocal characteristics using “MFCC” and a “BNN”. The cries were divided into three categories: hunger, discomfort, and tiredness. Prior to classification, the input audio was validated as an infant cry using three features: pitch, energy, and harmonic ratio. The MFCC features were then analyzed and classified using the BNN, resulting in an overall accuracy of 98%.

Sidiq M et al. [9] designed a system that accepted input sound signals. The feature extraction technique employed in this study were the MFCC and HMM. The initial stages of MFCC involved partitioning the voice signal into frames, which are then processed using the Mel-filter bank derived from the human auditory system. The dictionary encompassed 10 words, comprising 5-digit words and 5-command words. The recordings were obtained from 30 speakers, both male and female, with each word spoken 5 times, resulting in 150 sound files per word. The test results demonstrated that the system was capable of recognizing the mentioned words of speakers with an accuracy of “93.89%” in a noise-free environment, and “58.1%” in environments with noise.

Barua P et al. [10] developed a system to recognize speech from human voices using “MFCC” as features. They used “16” mel-scale warped cepstral coefficients to recognize speech from two Bangla language commands. The commands “turn on light” and “turn of fan” spoken by a specific individual were used to extract features for a neural network. The system achieved a speaker recognition rate of nearly 83% and a speech recognition accuracy of 60%.

CheYong Yeo et al. [11] proposed a method for animal identification based on distinct voice patterns. The approach utilized ZCR and MFCC for feature extraction. For classification, DTW was employed to distinguish between different voice signals. However, the paper did not provide specific details about the dataset used. The system achieved an accuracy of 75%

Molnar et al. [12] proposed method to discriminate various dog barks based on its voice. Mean of the first formant, STD of harmonics, RMS of spectral skewness, spectral rolloff, spectral derivative was used as features. Anova and Bayes models were used as classifiers. Dataset size was 6,646. Data not available to download in public platform. Accuracy of the system was measured to be 43 to 52%.

Table-1 below provides summary of related works carried out on the dog voice emotion data. Gap findings observed in the review papers are highlighted.

TABLE 1. SUMMARY OF THE LITERATURE SURVEY

Author	Dataset availability	Feature-extraction/Classifier	Accuracy/Gap Findings
<i>Ferres K(2022)</i>	Dataset size 13,809/Images)	Deeplabcut framework	Emotion was identified based on dog image posture. Dog Voice data was not considered
<i>Scoresby(2021)</i>	Not applicable	Review paper based on positive impact on human mental health by inclusion of domestic pets	Not applicable
<i>Cleary Michelle(2021)</i>	Not applicable	Review paper based on the cruelty of pet owners against the subject	Not applicable
<i>Schnider(2020)</i>	1200 hours of voice recordings which includes dog and other sounds	LDA was used reduce the dimensionality from 8239 features to 51 features. SVM, Adaboost, Random forest etc.	False positive rate 2% and Sensitivity 89%. Single dog bark recordings and Binary classifier problem statement.
<i>Sibiryakova(2020)</i>	Dataset size 824, publicly not available	Features: FFT tool to compute frequency content in dog-whines	The effectiveness of the proposed model was not evaluated, it focused on analysis of dog-whines rather than classification
<i>Rosita D(2016)</i>	Dataset size 180(positive) and 25(negative). Publicly not available.	Feature extraction: MFCC, Pitch, Autocorrelation Classifier: Back propagation Neural network	Accuracy achieved on the Dataset was 98%. Author had considered infant cry voices. Dataset was unbalanced, positive classes were more in numbers compared to negative classes.
<i>Siddiq M(2016)</i>	150 recordings of each word from 30 subjects(Men+Women). Dataset not available publicly	Feature extraction: MFCC, Feature modelling: Hidden Markov Model.	Accuracy under noisy conditions was monitored to be 58%. Human voice was used for recordings.
<i>Che yeong(2012)</i>	Not mentioned in the paper. Very few samples of voice considered	Feature extraction: MFCC, ZCR Classifier: DTW	Author has considered very few samples of voice for the proposed model and accuracy was recorded to be 75%
<i>Molnar(2008)</i>	Dataset size 6646. Not available in public domain	Features: Mean of the first Formant, Standard deviation of harmonics, Classifier: Anova and Bayes classifiers	Model effectiveness was reasonably low. Accuracy was monitored to be around 43 to 52%.

III. DOG EMOTION DATASET

In this section, we focus on dog emotion understanding using various dog bark types.

A. Dog bark emotion dataset

The dataset used for designing emotion identification system includes 4 classes, namely other animal bark, normal dog bark, dog grunting, dog growling. Dataset consists of 163 audio files, training data set size 115 files and testing data set size 48 files. Other animal bark training data size 36, testing data size 14. Normal bark training data size 33, testing data size 13. Dog grunt training size 23, testing size 11. Dog growl training data size 23, testing data size 10. Other animal bark(cat)and few samples of normal dog bark are picked from the open-source cat-dog audio Kaggle dataset, these signals are sampled at sampling frequency of 16000 Hz [13]. Dog grunt and dog growl voices are acquired from the subject (1.5-year-old female dog having chronic diarrhea syndrome) using voice recorder app sampled at 44.1KHz. The dataset DEMO-lite(Dog emotion-lite) is available to be downloaded for free [14].

Fig.1.a depicts the time versus frequency information interpretation of the other animal voice recorded with 5 seconds duration sampled at 16 kHz.

Fig. 1.b. shows the time versus frequency information captured for the normal dog bark voice sampled at 16kHz and recorded with 8 seconds long duration.

Fig. 1.c. displays the time versus frequency information analyzed of the dog grunt voice decimated by 3 factor and recorded with 5.68 seconds duration.

Fig. 1.d. illustrates the time versus frequency information visualization of the dog growl voice decimated by 3 and recorded with 4 seconds duration.

B. Spectrogram analysis of Dog barks

To analyse the signal both in time and frequency, spectrogram analysis can be employed. Spectrogram graphs provide frequency content of the signal over time, pixel intensity in the colormap of the spectrogram provides amplitude of the signal at that time and frequency. In the context of a spectrogram, the dB scale is used to represent the magnitude of the signal in each time-frequency bin. A higher dB value indicates a stronger signal component.

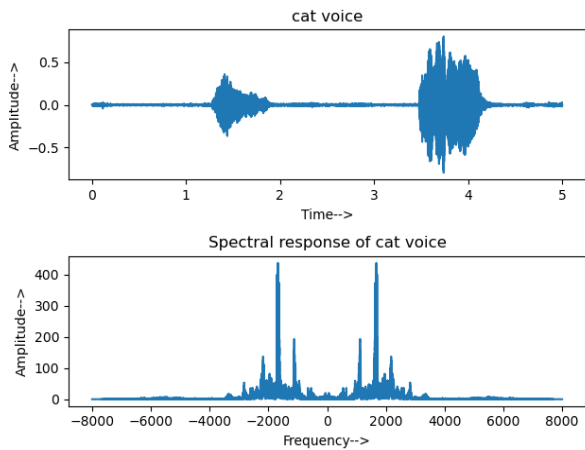


Fig. 1. (a). Time vs Frequency response of other animal voice

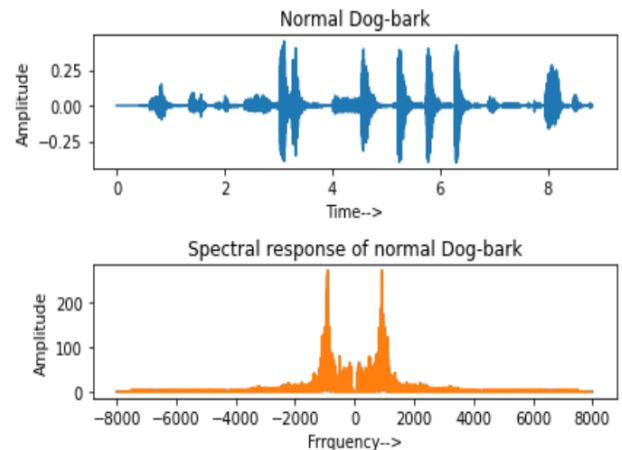


Fig. 1. (b). Time vs Frequency response of normal dog bark

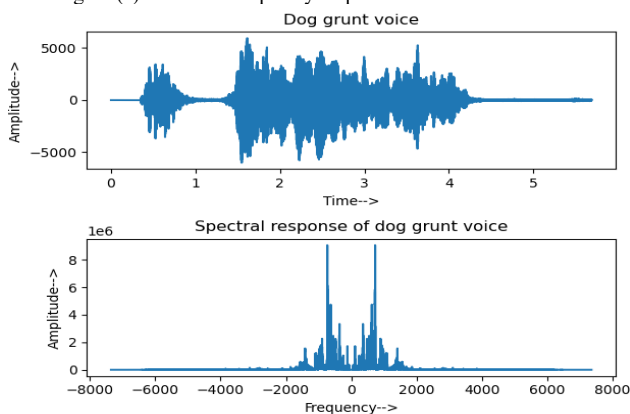


Fig. 1. (c). Time vs Frequency response of dog grunt voice

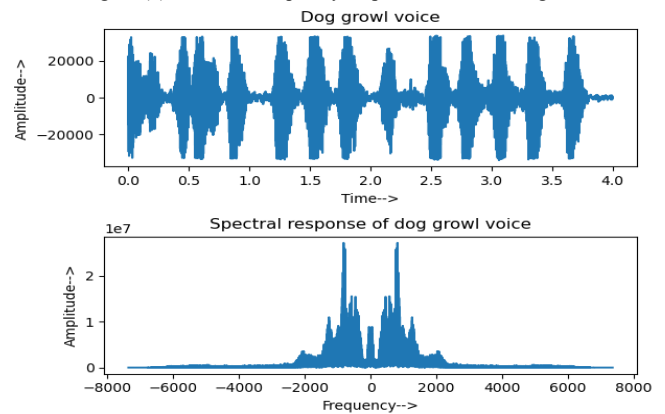


Fig. 1. (d). Time vs Frequency response of dog growl voice

Fig. 1. Time vs Frequency response of various types of dog-emotion

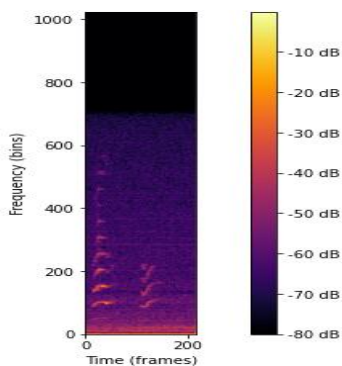


Fig. 2. (a). Spectrogram analysis of other animal voice

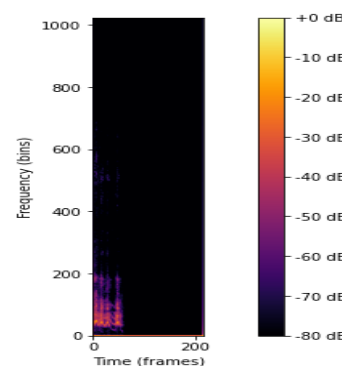


Fig. 2. (b). Spectrogram analysis of Dog bark voice

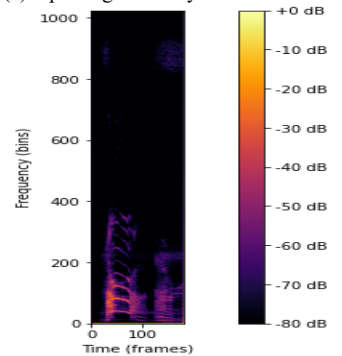


Fig. 2. (c). Spectrogram analysis of Dog grunt voice

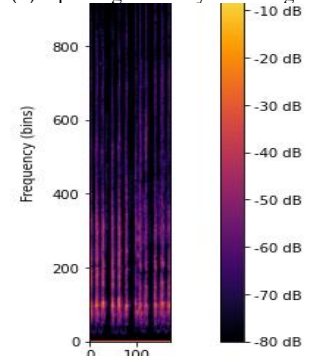


Fig. 2.(d). Spectrogram analysis of Dog growl voice

Fig. 2. Spectrogram response of various types of dog-emotions

The Figures. (2.a - 2.d) shows the spectrogram of various forms of dog barks. The visualization and interpretation plots of time vs frequency analysis and spectrogram were captured with the help of online virtual laboratory simulator

which utilizes numpy and matplotlib packages running in the background and thus furnishing the details on the html page[15].

C. Challenges in classifying Dog barks

The dataset comprises dog vocalizations sampled at two distinct frequencies: 44kHz and 16kHz. Prior to performing feature extraction, it is essential to normalize these sampling rates. An analysis of time-domain signals sampled at 48kHz revealed, through spectral inspection, that no significant energy existed above 3.5kHz. Consequently, the signals were down sampled by a factor of 3, effectively reducing the sampling rate to approximately 16kHz.

In the “dog grunt” category, many segments contain silence. When these frames are subjected to logarithmic transformations, errors such as negative infinity may arise due to $\log(0)$. To mitigate this, small non-zero values can be added to such silent frames.

From Figures 1.a to 1.d, it is evident that the dominant frequency components remain below 3.5kHz. Therefore, down sampling the input followed by the application of an IIR low-pass filter is an effective preprocessing step for improved signal analysis.

IV. METHODOLOGY

In this section, feature extraction and classification algorithm for dog voice identification is discussed.

A. Proposed Algorithm

Step 1: Collect dog voice recordings using a suitable audio recording application. These recordings make up the dataset.

Step 2: Split the dataset into training and testing sets. In this case, the dataset contains 163 dog or animal voice samples, divided into 115 for training and 48 for testing.

Step 3: Apply Wavelet/LMS based denoising to filter non stationary noises. Examine the sampling frequency of each audio file. If any file has a sampling rate below 14kHz or above 16kHz, it needs to be resampled. In this dataset, around 66 audio samples were originally recorded at 44.1kHz, so they were down sampled to 14.7kHz. The rest of the audio files were already at 16kHz and were therefore left unchanged [16].

Step 4: Define the filter design parameters—passband frequency, stopband frequency, passband attenuation, stopband attenuation, and sampling frequency—and apply an IIR low-pass filter using the Butterworth design method to the emotional voice signals. For this application, the parameters were set as: Passband(PB) = 3000 Hz, Stopband (SB) = 4000 Hz, Passband Attenuation(A_p) = 1 dB, Stopband Attenuation(A_s) = 20 dB, and Sampling Frequency(F_s) = 16000 Hz. These values were chosen based on the observed spectral characteristics in Figures 1.a–1.d.[15,16].

Step 5: Using the chosen specifications, compute the filter coefficients—numerator coefficients (b’s) representing zeros and denominator coefficients (a’s) representing poles. Then, apply the filter to the audio signals using these coefficients.

Step 6: Extract Mel-Frequency Cepstral Coefficients (MFCCs) from the filtered audio signals. These MFCCs serve as the key features to be stored in the database for further analysis or classification tasks [15].

Step 7: Choose a suitable classifier for the task—in this case, a Deep Neural Network (DNN)—to perform effective

classification of the extracted features. The DNN is trained over 50 epochs, and the resulting weight matrix is saved for later model evaluation. Both the training accuracy, validation accuracy and the value of the loss functions at the end of the 50th epoch is recorded.

Step 8: The trained model, using the weight matrix obtained in Step 7, is then applied to the testing dataset. The testing accuracy and corresponding loss value are evaluated and recorded to assess the model’s generalization performance.

Steps 1-2 constitute the data acquisition and management phase, where the dog voice samples are collected and divided into training and testing datasets.

Steps 3–5 cover preprocessing, including denoising methods, sampling rate normalization, IIR Butterworth filtering, and MFCC feature extraction.

Steps 6–8 involve Feature extraction and Deep Learning model selection with varying hyper parameters.

A flowchart illustrating the overall process of data acquisition, preprocessing, and feature extraction is provided in Fig. 3. Suitable audio-recorder App was used to record the emotion data. The data collected were pre-processed using Wavelet- IIR or LMS-IIR based hybrid model. Filtered data was applied on the MFCC algorithm, which serves as the features to the classifier.

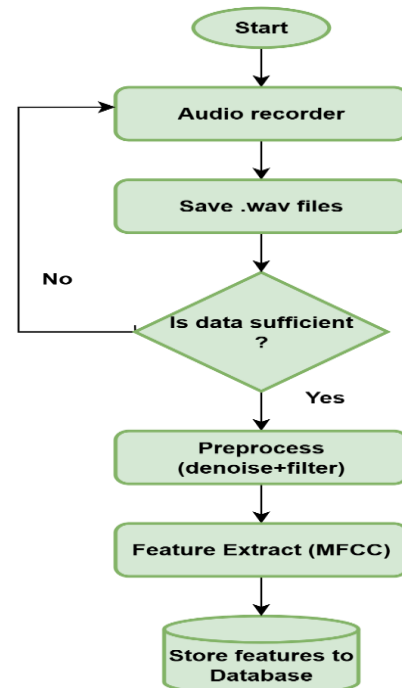


Fig. 3. Flowchart depicting the Data acquisition, Preprocessing and Feature extraction of dog emotion recognition model

The flowchart of dog emotion classification model is shown in Fig. 4. Stored MFCC coefficients serves as the input to the various Deep Learning models. The selection of cost function and optimizer is based on the performance produced by the model. The model performance on sparse categorical cost function and adam optimizer were optimal. Other alternative cost function used in the approach included cross-entropy, mean squared error, Hinge loss etc. As far as optimizers are concerned Gradient Descent, SGD, RMSprop etc. were used as an alternative.

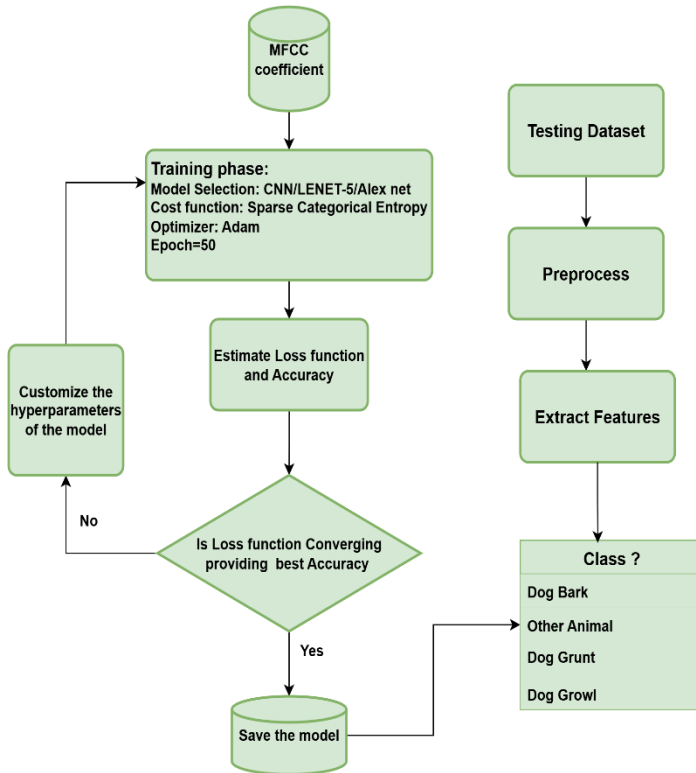


Fig. 4. Flowchart of dog emotion classification model using various CNN's

B. Preprocessing: Denoising

Wavelet based denoising and LMS adaptive filtering are employed on the noisy data. Fig. 5 provides the block diagram of the denoising models. The parameter configuration of wavelet denoising scheme is chosen as follows: Mother wavelet: **Daubechies-4(db-4)**, level of decomposition: **4**, threshold selection: **Universal threshold rule**.

First level decimated approximation version of signal $x(n)$ using db-4 wavelet is given by

$$y_{lp}(n) = (x(n) * lp(n)) \downarrow 2 \quad \dots(1)$$

Where $lp(n) = (0.378, 0.23, -0.129, -0.06)$ are db-4 low pass filter coefficients.

First level decimated detailed version of signal $x(n)$ using db-4 high pass filter coefficients is given by

$$y_{hp}(n) = (x(n) * hp(n)) \downarrow 2 \quad \dots(2)$$

Where $hp(n) = (-0.06586, 0.129, -0.23, 0.378)$ are db-4 high pass filter coefficients.

Configuration parameters of Adaptive LMS filtering scheme follows, $M(\text{Order})$: **16**, $\mu(\text{learning rate})$: **0.01**, Noise distribution: Gaussian noise with **zero mean** and **0.05 variance**.

The response of the LMS system with linear combination of inputs and weights is given by

$$y_{lms}(n) = w^T(n) x(n) \quad \dots(3)$$

The LMS weight update rule is defined as follows

$$w(n+1) = w(n) + \mu e(n) x(n) \quad \dots(4)$$

Where μ is step size and $e(n)$ is error defined as the difference between desired signal $d(n)$ and filtered signal $y(n)$.

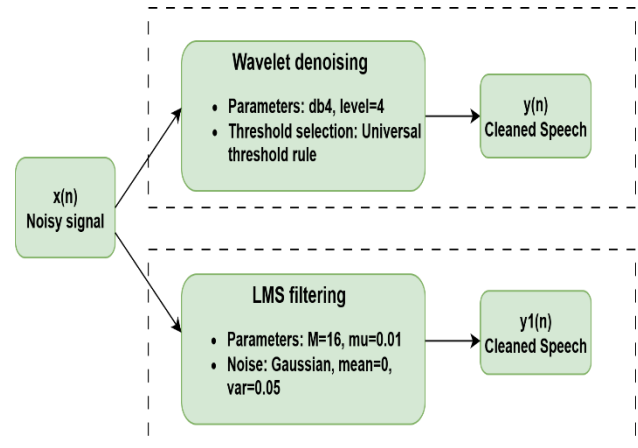


Fig. 5. Noise removal models

C. Preprocessing: IIR Butterworth filter

Low pass filter coefficients for passband frequency 3kHz, stopband frequency 4kHz, Sampling frequency 16kHz, passband attenuation 1dB, stopband attenuation 20 dB using bilinear transformation is given below.

$$b = [0.002, 0.018, 0.063, 0.127, 0.159, 0.127, 0.063, 0.018, 0.002]$$

$$a = [1, -1.587, 2.08, -1.53, 0.87, -0.32, 0.082, -0.012, 0.00085]$$

The transfer function for the above filter coefficients is defined as follows.

$$H(z) = \frac{\sum_{k=0}^8 b_k z^{-k}}{1 + \sum_{k=1}^8 a_k z^{-k}} \quad \dots(5)$$

After the process of pre-processing, MFCC coefficients from the speech signal is extracted for better analysis and classification [17].

D. Alexnet CNN design metrics

Model summary and architecture of the Alex-net CNN and its hyperparameters are discussed in this section.

MFCC coefficients are reshaped to 94x93 samples which acts inputs to Alex net model. Alexnet model uses five Convolutional block, three max pooling blocks and three fully connected dense neural networks. Alex net model summary generated using TensorFlow python package is depicted in Fig. 6. below [19].

Model: "sequential"		
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 84, 83, 96)	11712
max_pooling2d (MaxPooling2D)	(None, 41, 41, 96)	0
conv2d_1 (Conv2D)	(None, 37, 37, 256)	614656
max_pooling2d_1 (MaxPooling2D)	(None, 18, 18, 256)	0
conv2d_2 (Conv2D)	(None, 16, 16, 384)	885120
conv2d_3 (Conv2D)	(None, 14, 14, 384)	1327488
conv2d_4 (Conv2D)	(None, 12, 12, 256)	884992
max_pooling2d_2 (MaxPooling2D)	(None, 5, 5, 256)	0
flatten (Flatten)	(None, 6400)	0
dense (Dense)	(None, 4096)	26218496
dense_1 (Dense)	(None, 4096)	16781312
dense_2 (Dense)	(None, 4)	16388
Total params: 46740164 (178.30 MB)		
Trainable params: 46740164 (178.30 MB)		
Non-trainable params: 0 (0.00 Byte)		

Fig. 6. Model summary of the Deep Neural Network (Alex-net) model

Alexnet CNN architecture used for classification of pet voice is shown below in Fig. 7. [18,30].

Alex net model discussed in the above section utilizes the following parameters for estimating the model performance.

Convolution layer

Convolution layer forward propagation expression is given by

$$y_{ij,k} = \sum_{m=0}^{K-1} \sum_{n=0}^{K-1} \sum_{c=0}^{C-1} x_{i+m,j+n,c} \cdot w_{m,n,c,k} + b_k \quad \dots (6)$$

Where K is kernel, C is number of channels, w is filter and b is bias.

Max pooling layer

Max pool layer reduces the dimension of the input by retaining important information. The relationship between input and output upon performing max pooling operation is given by

$$y_{ij,k} = \max_{m,n} x_{i+m,j+n,k} \quad \dots (7)$$

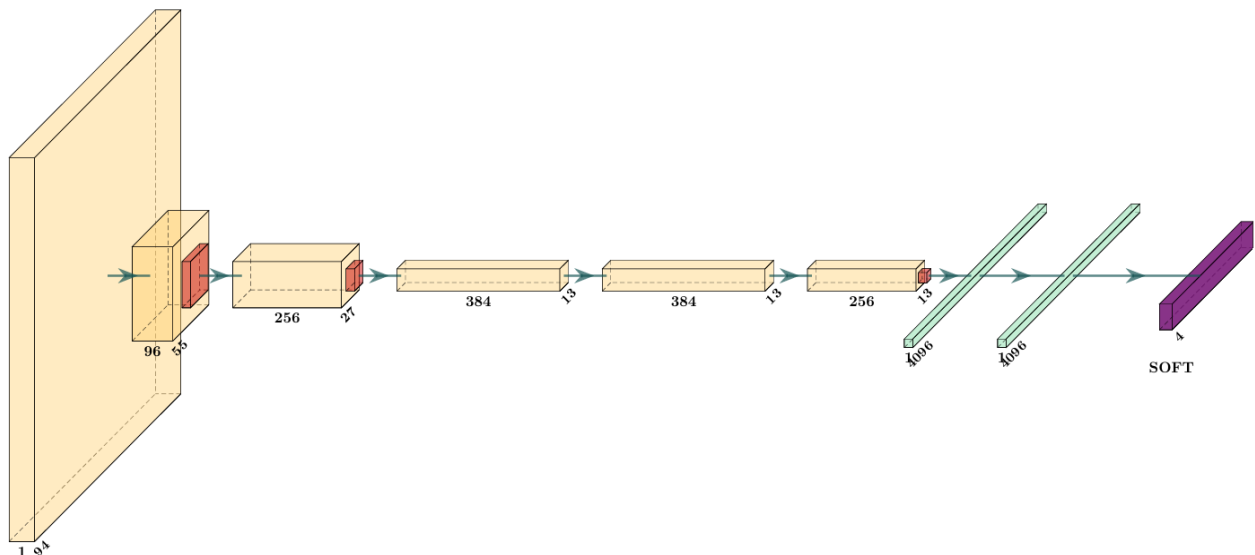


Fig. 7. Alex net CNN network model for pet emotion classification

Fully connected layer

After flattening the output obtained from the convolution-max pool layer, thus obtained data is forwarded to the fully connected layer. The expression used at the output of the fully connected layer is given by

$$y = W \cdot x + b \quad \dots (8)$$

ReLU activation-function

ReLU-activation function is used at the output of every convolutional and fully connected layers. Relu activation function for an input x is given by

$$f(x) = \max(0, x) \quad \dots (9)$$

Softmax activation-function

Softmax activation function is used at the output layer. For an input y_i it is given by

$$\sigma(y_i) = \frac{e^{y_i}}{\sum_{j=1}^k e^{y_j}} \quad \dots (10)$$

Where k denotes number of classes in the given problem statement.

Cross-entropy loss for true and estimated value of target function is given by

$$L = - \sum_{i=1}^k t_i \log(\sigma(y_i)) \quad \dots (11)$$

Where k denotes number of classes in the given problem statement [20].

Adam optimizer weight updating rule is given by

$$w_t = w_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \quad \dots (12)$$

Here η is the step size/learning rate, selected around $1e-3$, $\hat{m}_t$ is the bias-corrected first moment estimate and $\sqrt{\hat{v}_t}$ is square root of bias-corrected second moment estimate obtained from gradients of loss function [21].

V. RESULTS AND DISCUSSION

This section deals with; experimental results obtained for the dog-voice dataset.

A. VAD and SNR

To remove effectively the presence of noise and evaluate the signal quality, Voice activity detection system (VAD) and SNR scores are employed. Voice activity detection is determined based on adaptive heuristic approach, which separates voice from the speech depending on the threshold value set. Threshold value is determined by finding the median value on small chunks of the original data with 2ms duration.

$$T_h = \text{med}(x_{\text{frame}}) * 1.5 \quad \dots(13)$$

The median value produces robust estimate of the noise level and is less sensitive to outliers. Scaling factor of 1.5 is multiplied to boost the threshold to filter only those frames with higher energy as speech. Finally, SNR in dB is evaluated after separating the noise and speech frames using formula below.

$$\text{Snr}_{\text{dB}} = 10 * \frac{\text{Signal power}}{\text{Noise power}} \quad \dots(14)$$

Table-2 below provides the average SNR in dB evaluated on the other-animal, dog-bark, dog-grunt and dog-growl emotion classes with various noise-removal models. Overall, SNR scores produced by LMS+IIR preprocessing noise removal model was better compared to other noise filtering models, which includes No pre-processing, IIR-preprocessing and Wavelet+IIR preprocessing. SNR values recorded by LMS+IIR model on the other animal, dog-bark, dog-grunt and dog-growl training dataset was found to be 22.98, 29.90, 25.35 and 22.13 dB respectively. IIR preprocessing and Wavelet+IIR preprocessing showed good results on the dog-grunt dataset with SNR of 25.70 dB.

TABLE 2. SNR in dB evaluated for various noise filtering models on dog-emotion dataset

Noise Models	SNR in dB Other animal	SNR in dB Dogbark	SNR in dB Doggrunt	SNR in dB Doggrowl
No	19.90	27.66	25.44	20.59
preprocessing IIR based	19.85	27.70	25.70	20.59
preprocessing Wavelet+ IIR	19.85	27.70	25.70	20.6
preprocessing LMS+IIR	22.98	29.90	25.35	22.13
preprocessing				

B. ML- Classifiers using Sklearn Python package

Table-3 provides comparative study of various machine learning classifiers with customized model parameters applied on the demo-lite dataset without and with data preprocessing [22].

ML classifiers included Decision Tree, Random Forest, Support Vector Machine, Logistic Regression, K-Nearest Neighbor, Linear Discriminant Analysis, Gradient Boost Algorithm, Multi-layered perceptron(MLP). To compare the performance of various classifiers on the dataset hybrid models such as IIR+MFCC, Wavelet +IIR+MFCC and LMS+IIR+MFCC are considered [6].

It is evident from the Table 3 multi-layered perceptron model (Shallow Neural network) and Random Forest ML classifiers performed well on the dog-emotion dataset. Random Forest classifier with Wavelet denoising and IIR filtering produced **testing accuracy of 73%**. Multi-layered perceptron model, recorded **accuracy of 71%** without and with pre-processing (IIR). Moreover, sklearn python package enabled with ML classifier had restrictions on setting loss functions and defining validation set. To fine tune the data with several configuration parameters and to make use of various deep learning classification models, accelerate the training speed (GPU-driven) tensorflow-gpu python package is employed for further classification analysis.

TABLE 3. ML classifiers applied on Dog-emotion dataset with and without Preprocessing

ML Classifier	Feature Extraction: MFCC		Feature Extraction: MFCC Preprocessing: IIR		Feature Extraction: MFCC Preprocessing: Wavelet+IIR		Feature Extraction: MFCC Preprocessing: LMS+IIR	
	Preprocessing: No							
	Train Acc	Testing Acc	Training Acc	Testing Acc	Training Acc	Testing Acc	Training Acc	Testing Acc
DT	92	72	87.82	64.5	88.69	68.75	71.3	53
SVM	86	58.33	87	62.5	87.28	64.6	85.217	61
LR	100	66.66	100	68.75	100	68.75	100	61
KNN	82.6	55	86	66.6	86.95	64.58	78.26	58.33
LDA	90.43	65	92.17	68.75	90.43	62.5	93.91	62.5
GBOOST	100	62.5	100	58.33	100	54.16	100	45.83
RF	91	70	93	70	97.39	73	92.17	56.25
MLP	100	71	100	71	100	66.67	100	61

C. Deep Learning- Classifiers with tensorflow-gpu

CPU specifications: Intel I-5 6th generation CPU with 4GB RAM running on Windows 10.

GPU specifications: Nvidia Geforce-920M 2GB of DDR3 memory with compute capability 3.1. Compatible Drivers for GPU: Python Interpreter-3.6, tensorflow- 2.0.0, CUDA driver version 10.0, CuDNN version 7.4.1.

Table-4 provides the performance comparison offered by the system with CPU and GPU. It can be observed that as the number of trainable parameters increased the performance offered by GPU was far optimal compared to CPU. When the number of trainable parameters was less as in the case of 5-DNN the performance of CPU and GPU were almost equal. Alex-net CNN Deep learning classifier which performed 5 times faster on GPU compared to CPU with a total trainable parameter of 46,740,164.

TABLE 4. CPU vs GPU performance measure on dog-emotion dataset

Classifier	Trainable parameters	Elapsed time on CPU with 50 Epoch(seconds)	Elapsed time on GPU with 50 Epoch(seconds)
5-DNN	1,533,540	4.71	5.31
CNN	66,10,628	66.13	29.95
Lenet-5	504,716	16.72	9.15
Alex-net	46,740,164	731.619	156.817
Resnet 50	24,053,892	1048.011	213.719
LSTM	28,709,956	100.5	45.3

The training dataset[14] is further splitted into the ratio of 70:30 to facilitate validation. Deep Learning Classifiers used for analyzing the emotion dataset included 4/5- layered

Deep Neural network, Convolution Neural network (3 convolution layers, 2 max pooling layer and 2 dense layers), Lenet-5 CNN, Alex-net CNN, LSTM, Resnet-50 and Vggnet-16. Table-5 below provides the validation accuracy/loss, validation accuracy/loss and testing accuracy/loss recorded using various Deep learning models. Models which performed consistently well on the validation dataset and testing dataset were identified. Alex-net CNN with IIR+MFCC, Lenet-5 CNN with IIR+MFCC filter, and CNN with Wavelet+IIR filter recorded good testing classification accuracy of **87.5%**, **85.4%** and **85.4%** respectively. Compared to the models proposed by the authors [6,9,11] the result obtained was quite encouraging. Other popular Deep-learning frameworks such as Resnet-50, Vggnet-16, Mobilenetv2 and Inceptionv3 worked well on the un-preprocessed data with highest accuracy recorded not exceeding 71%[28].

Deep learning classifiers such as Vggnet-16, Mobile net, Inception fared poorly on the filtered dataset as they are specifically used for Image Classification. Also, these classifiers relied heavily on the ImageNet pre-trained model.

D. Accuracy, Confusion matrix, Classification report and ROC

Fig. 8. Presents a bar graph comparing the training, validation, and testing accuracy of effective hybrid filter based deep learning classifiers identified from table-5. Various filtering-classifier selected for comparison includes, IIR-MFCC based 4-layered Deep Neural network (IM-4DNN), IIR-MFCC based 4-layered Deep Neural network (IM-4DNN), , IIR-MFCC based Alexnet-CNN(IM-Alexnet), IIR-MFCC based Lenet-5 CNN (IM-Lenet-5), LMS IIR-MFCC based CNN(LIM-CNN), MFCC based LSTM(M-LSTM) , Wavelet IIR MFCC based 4 layered Deep Neural network (WIM-4DNN) and Wavelet IIR MFCC based CNN(WIM-CNN). It is evident from the comparison graph, IM-Alexnet produced robust results on the dog-emotion dataset [27,28].

TABLE 5. Deep Learning classifiers applied on Dog-emotion dataset

Classifier	MFCC			MFCC+IIR			Wavelet+IIR+MFCC			LMS+IIR+MFCC		
	Train Acc	Valid Acc	Test Acc	Train Acc	Valid Acc	Test Acc	Train Acc	Valid Acc	Test Acc	Train Acc	Valid Acc	Test Acc
4-DNN	100	91.43	71	100	88.57	81.25	100	80	81.25	100	75	62.5
5-DNN	100	94.29	73	100	80	77	100	85.71	75	100	74.29	67
CNN	100	85.71	80	100	85.71	77.1	100	85.71	85.4	100	88.57	79
Lenet-5	100	85.71	79.1	100	91.43	85.41	100	97.14	80	100	85.71	72.9
Alexnet	98.75	85.71	65	100	88.57	87.5	100	82.86	77.1	97.5	72	67
LSTM	95	85.71	73	98.75	82.86	60.4	97.52	77.74	64.58	98.75	74	62.5
Resnet-50	96.25	77.14	56.2	97.5	74.29	58.33	97.5	65.71	60.42	97.5	62.5	50
Vggnet-16	91.25	71.43	70.8	88.75	74.29	56.25	85	80	65	92.5	74	62.5

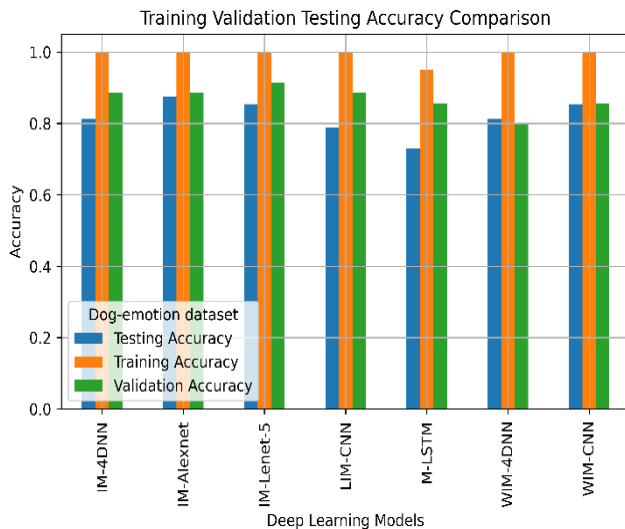


Fig. 8. Deep-learning models accuracy comparisons

Fig. 9. Illustrates the training and validation loss functions, recorded for the AlexNet CNN with IIR+MFCC filter, Lenet-5 CNN with IIR-MFCC filter and CNN with Wavelet-IIR-MFCC filter considering various epoch values. Solid lines in the figure indicates training losses and dashed lines indicates validation loss function values. The epoch values selected for this experimental setup were 20, 30, 40, and 50 respectively. Training loss function values for the above mentioned three models were optimal at Epoch: 40. Validation loss function deviated initially for lower epoch's, but stayed stable after increasing the epoch beyond 50[23,24].

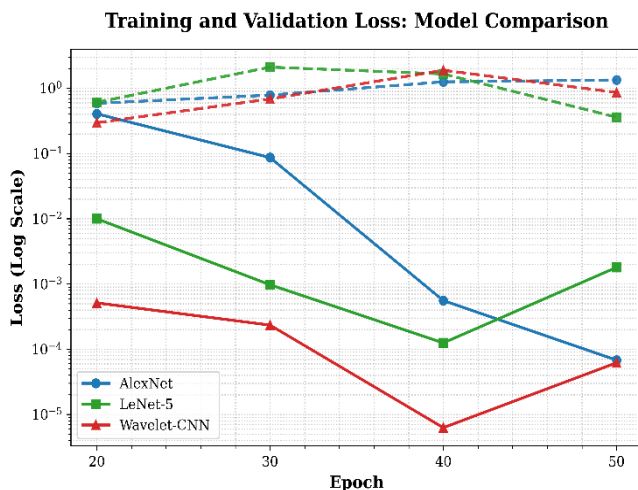


Fig. 9. Deep Learning model Loss vs Epochs

Fig. 10. depicts the confusion matrix results for the AlexNet with IIR+MFCC filter deep learning model. The evaluation was conducted on a testing dataset comprising 48 samples. The "other animal bark" class included 14 instances, of which 10 were correctly classified and 4 were misclassified. For the "dog bark" class, 12 out of 13 samples were accurately identified, with 1 misclassification. In the case of the "dog growl" class, 10 out of 11 samples were correctly classified. The "dog grunt" class achieved perfect classification, with all 10 samples correctly identified.

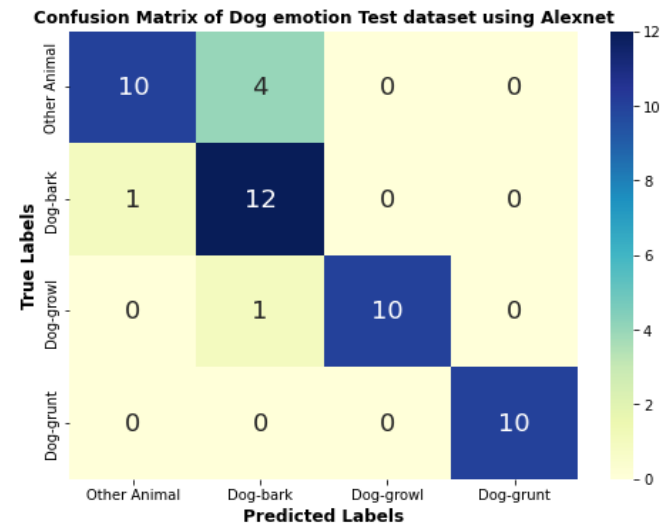


Fig. 10. Confusion matrix of dog voice testing dataset using Alex-net CNN

Fig. 11. presents the classification performance metrics for the AlexNet with IIR+MFCC filter. Precision, Recall and F1- Score for the Dog-emotion dataset is listed below [23, 24,25,26].

Classification Report with Alexnet IIR-MFCC			
	precision	recall	f1-score
OtherAnimal	0.91	0.71	0.80
Dog-bark	0.71	0.92	0.80
Dog-grunt	1.00	0.91	0.95
Dog-growl	1.00	1.00	1.00
accuracy	0.88	0.88	0.88
macro avg	0.90	0.89	0.89
weighted avg	0.89	0.88	0.88

Fig. 11. Classification report of dog voice testing dataset using Alex-net CNN

To evaluate the model classification performance on the dog-emotion dataset, ROC-AUC curve design metrics is employed. ROC-AUC curve of Alex-net model on the dog-emotion dataset is shown below in Fig. 12. [25,26].

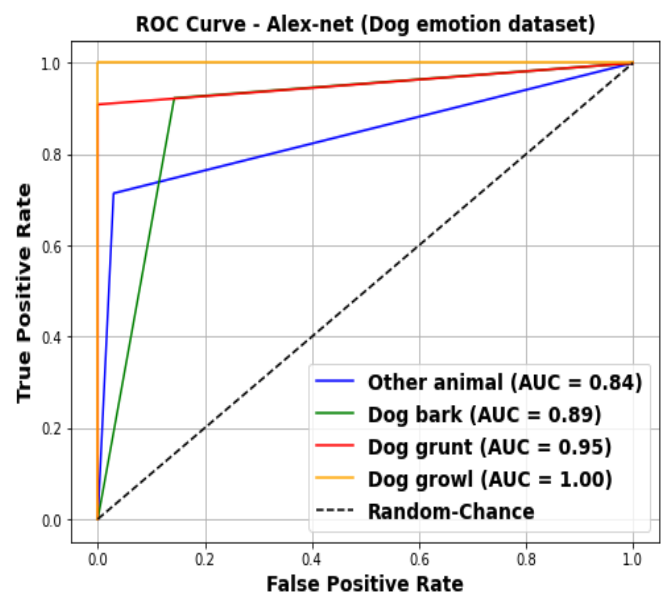


Fig. 12. ROC-AUC curve of Alex-net model

TABLE 6. Deep Learning classifiers applied on Cat vs Dog Audio Dataset

Classifier	MFCC			MFCC+IIR			Wavelet+IIR+MFCC			LMS+IIR+MFCC		
	Train acc	Valid acc	Test acc	Train acc	Valid acc	Test acc	Train acc	Valid acc	Test acc	Train acc	Valid acc	Test acc
<i>CNN</i>	100	96.43	98	100	95.24	97.62	100	96.83	97.14	100	95.63	96.19
<i>Lenet-5</i>	100	94.84	95.71	100	95.63	95.2	100	94.44	93.33	100	93.65	95.74
<i>Alexnet</i>	95.41	87.7	86.67	95.75	90	92	93	85.71	91	94.56	87	92.3
<i>5-DNN</i>	100	94.44	95.23	100	93.25	93	100	93	94.76	100	92	91
<i>4-DNN</i>	100	93.25	93.33	100	92.46	94	100	91.27	93.33	100	87.3	89

E. Cat vs Dog dataset

This section deals with application of Hybrid models and Deep Learning classifiers on cat versus dog voice dataset [29]. The dataset includes 566 cat and 484 dog voice recordings sampled at uniform sampling frequency of 16kHz with two channels and variable recording size. Total number of samples selected for training from the cat and dog voice samples were 452 and 388. Remaining samples were considered for testing. Table-6 above depicts training, validation and testing accuracy recorded for the cat vs dog audio dataset. Five best classifiers which performed well on the dog-emotion dataset are considered with four noise filtering approach as discussed in the earlier section. All the classifiers performed well on the dataset with validation and testing accuracy of above 85%. CNN model with 3 convolutional layer(3C) and 2 max pooling layers(2M) recorded the best performance on all four hybrid noise filtering models with an average validation and testing accuracy of 96% and 97.2%. Lenet-5 CNN model recorded the second-best performance with an average validation and testing accuracy of 94.64% and 94.99%. Average accuracy recorded in [31] was 90% which was relatively less compared to the proposed four approaches.

Training and Validation Loss: Model Comparison

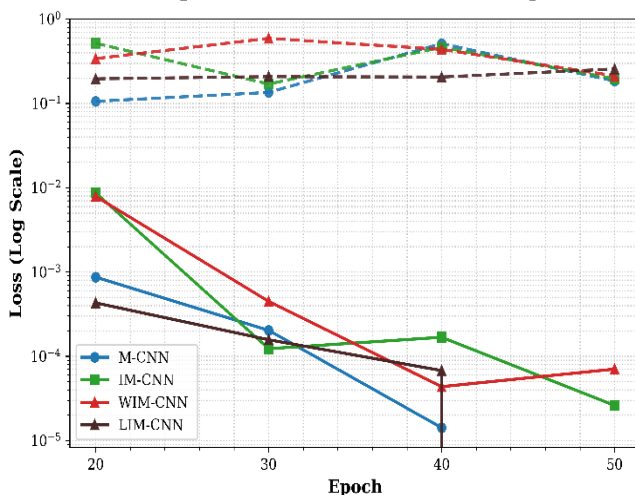


Fig. 13. Hybrid CNN model Loss vs Epochs

Fig. 13. above shows training and validation loss recorded on the cat vs dog audio dataset with models M-CNN, IM-CNN, WIM-CNN and LIM-CNN models. As the numerical loss function values were relatively low, logarithmic scale is adapted to discriminate the values of loss function for various models. The dotted line indicates validation loss and

solid lines indicate training loss. The various values of Epoch selected for the conduction of the experiment are 20,30,40 and 50 respectively. M-CNN model provided lowest value of training loss and validation loss function which were measured as 1.18e-5 and 0.1058. Similarly, IM-CNN produced largest value of the training loss of 8.7e-3 and WIM-CNN produced largest value of the validation loss of 0.5913 respectively.

VI. CONCLUSION

The paper aimed at developing an automated voice recognition and classification systems for domestic pets. Paper suggested some of the challenges encountered while recording pet voices which includes voices recorded with different sampling frequency, noise with speech, silence detection etc. Handling signals sampled at different sampling frequency is achieved by decimation. Removal of noises and unwanted frequencies is achieved through IIR LP filtering, wavelet denoising and LMS adaptive noise filtering-based approach. The proposed system also utilized an online virtual laboratory simulator to perform analyses such as time-domain versus frequency-domain response, Spectrogram data interpretation, IIR filter design, and MFCC-based feature extraction. Experimental results proved that IIR MFCC based Alex-net CNN, IIR MFCC based Lenet-5 CNN and Wavelet IIR MFCC based CNN produced good results on the dog-emotion testing data. Deep Learning models worked really well on the pre-processed dog-emotion dataset.

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